

Asael Acosta

Boston, Massachusetts | (626) 863-5420 | asael906@mit.edu | linkedin.com/in/asael-acosta

EDUCATION

Massachusetts Institute of Technology, Cambridge, Massachusetts

May 2028

Candidate for B.S. in Aerospace Engineering

GPA: 4.4 / 5.0

Coursework: Materials and Structures, Signals and Systems, Thermodynamics, Fluid Dynamics, Astrodynamics, Space Propulsion, Satellite Engineering

SKILLS

Programming: Python (PyTorch, Orekit, SciPy), LaTeX, Bash, Java (WPILib), /HTML/CSS/ JS (jQuery), MATLAB

Software: QGIS, Ansys STK (Astrogator), GMAT, Solidworks, nTop, Git, Solidworks, Miro, Microsoft Suite,

Languages: English (native), Spanish (fluent)

EXPERIENCE

MIT Satellite Club, *Treasurer, Communications Subteam*, Cambridge, MA

October 2025 - Present

- Budgeting and procuring funds for parabolic flight demonstration of cryogenic boiloff-reduction payload and remote-sensing cubesat launch
- Developing communications schemes and technical documentation for payload monitoring
- Building tutorials in orbital mechanics and relevant software tools (e.g., STK and Orekit) to train new members
- Presenting Concept of Operations, Preliminary Design Review to industry representatives to checkpoint progress

MITRE Corporation, *Systems/Digital Engineering Intern* Bedford, MA

June - August 2025

- Developing systems analysis tool for government sponsor, leveraging git for collaboration
- Meeting regularly with sponsor representatives to clarify project requirements and add features
- Developing project management tool using Microsoft Suite to streamline collaboration between sponsor and John Hopkins Applied Physics Laboratory
- Outlining data pipelines between information sources and displays to increase productivity metrics

MIT Varon Lab, *Undergraduate Researcher*, Cambridge, MA

September 2025 - May 2026

- Trained U-Net model to predict methane-sensitive bands from non-sensitive satellite imagery to improve methane point-source and plume detection
- Collected and preprocessed ABI GOES data for training, validation, and testing datasets
- Investigated novel ML approaches used by the group and presenting progress at weekly meetings

National Polytechnic Institute, *AI Research Intern*, Querétaro, Mexico

May 2025 - August 2025

- Integrated Google DeepMind's Gencast forecasting model into a locally hosted hurricane trajectory predictor and visualization tool, reducing blurring from its previous iteration
- Translated and summarized ~20 English-written papers on AI weather forecasting techniques onto Spanish LaTeX document for future work on forecast downscaling
- Presented research to CENAPRED, the Mexican disaster prevention and response bureau, and to Google DeepMind representative in discussing further collaboration in their disaster prevention and mitigation initiatives

PROJECTS

NASA RASC-AL Challenge (MELIORA), *Concept of Operations*, Cambridge, MA

October 2025 - Present

- Designed system architecture for communication, navigation, positioning, and timing system to support long-term scientific missions to Mars
- Started with stakeholders to iteratively upscale detail on system specifications in order to define research direction
- Authored technical document outlining trade space analysis and technical justification for system as submission to nationwide competition to become a finalist
- Presented revised analysis to industry experts, including NASA and Northrop representatives, to achieve best in category award, 2nd place overall, and a reserved spot to present at SciTech 2027

Astrodynamics Final Project, *SSO Mission Planning and Analysis*, Cambridge, MA

February - May 2026

- Simulated SSO orbit of CubeSat fleet under various degrees of perturbation using Orekit to demonstrate orbital data center formation flight concept
- Reviewed simulation results and existing literature to select thrust maneuver strategies to maximize mission duration and gracefulness of degradation
- Presented findings to faculty, NASA JPL Engineers, FFRDC researchers to receive A+ for the project and course